# Arrival · Supported

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

**1. Fraction**

What percentage is 1/4?

Help: Read or point first. Multiply by 100.

**2. Genotypes**

What is the genotype ratio for Aa × Aa?

Help: Read or point first. Count AA, Aa and aa.

**3. Phenotypes**

With complete dominance, what is the phenotype ratio?

Help: Read or point first. AA and Aa show the dominant phenotype.

**4. Independence**

Does an aa offspring force the next offspring to be AA?

Help: Read or point first. Each fertilisation is a new event.

My response: _____

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# Independent · Supported

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

**1. Calculate**

Complete the expected aa counts for A, B and C.

Help: For each sample, write total ___. Find one quarter of that total for the expected aa count.

**2. Convert**

Convert observed 7/20 and 18/80 to percentages.

Help: Write two calculations: 7 ÷ 20 × 100 and 18 ÷ 80 × 100. Keep the correct total each time.

**3. Explain**

Explain why 7/20 does not automatically disprove the one-in-four model.

Help: Compare predicted probability with this small sample. Use: expected ___, observed ___; chance can produce ___.

My response: _____

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

Choose one route for the full 10 minutes.

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# Exit ticket · Supported

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

### 1. Calculate

Cross Aa × Aa. Find the expected aa count in 80 offspring.

Help: Find one quarter of 80. Write: working ___; answer ___; unit or percent ___.

### 2. Interpret

Why might a sample of 20 contain 7 aa rather than exactly 5?

Help: Use: my answer is ___ because ___. Point to a model or card first. Small samples vary by chance.

My response: _____

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# Expected versus observed

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

All samples model Aa × Aa. Calculate the expected aa count, then compare it with the simulated result. Do not change the theoretical probability because of one sample.

| Sample | Total | Expected aa | Observed aa |
|---|---|---|---|
| A | 20 | ___ | 7 |
| B | 20 | ___ | 3 |
| C | 80 | ___ | 18 |

**My evidence and reasoning**

**One limit or useful check**

My response: _____

Keep your first answer. Hear the feedback. Choose one real change, or explain why to keep it. Ask for a read-back if it helps.



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# Punnett square · working sheet

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Record the parent alleles. Fill all four boxes. Then find the chance of aa.

| **Aa × Aa · side / top** | **A** | **a** |
|---|---|---|
| A | _____ | _____ |
| a | _____ | _____ |

Compare with Aa × aa.

| **Aa × aa · side / top** | **A** | **a** |
|---|---|---|
| a | _____ | _____ |
| a | _____ | _____ |

Aa × Aa chance of aa: _____ . Aa × aa chance of aa: _____ .

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